Problem Set 11 Discussion

25 November 2025

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Announcements

- Last class session!
  - Complete course evaluations by Friday if you haven't already
- Make sure all your data lunch reflections are uploaded by **Dec 5th**
  - Can be in a separate document
- All problem set grades will be out by Friday – check Canvas
  - Learning notebook and data lunch grades will be out after **Dec 5th**
  - Final project feedback will be uploaded to your repositories week of **Dec 15th**
- Upload your **learning notebook** to the GitHub by **Dec 1st! (This Sunday)**

Learning notebooks

- Check example under Grading on Wiki
- Should be a fairly informal collection of your personal notes / takeaways / useful commands over the semester
- Final notes document should be formatted as Quarto markdown
  - Can copy paste from your preferred note format
- R-specific examples:
  - Tips for navigating RStudio
  - Definitions of useful functions you don't want to forget
  - Specific functions that were helpful for your final project
  - Packages discussed that you could use for your research
  - Lines of code you found / wrote that you want to save
  - Short examples of tricky concepts

Jupyter Notebook Tips

- **Kernel** tab - how to interrupt or clear output of kernel
- **Edit** tab - clear output
- **Run** tab - render all markdown cells
- **Help tab** - can access Python documentation refernce
  - **Show contextual help** - side panel with help documentation when clicking on functions

Constraints

SQL is good for data integrity

- generally want to define constraints at table creation
- sometimes can add later but may complain

- UNIQUE = every value has to be distinct
- NOT NULL = not a null value
- PRIMARY KEY = has both NOT NULL and UNIQUE, can only have 1
- DEFAULT = assign a default value if not given
  - call default value by not including anything in entry

- CHECK = value has to meet certain conditions to exist in the column
  - name cannot belong to a male
  - must belong to a set
  - value must be higher/lower

Reversing Strings

```
# no built in function, use reverse slicing..
s = "It is a truth universally acknowledged..."
s[::-1]
```

```
'..galP htraD fo dnegel eht uoy dlot reve I evaH'
```

R Recycling

```
x <- c(1,2,3)
y <- c(3,4,5)
x+y = (4,6,8)

# one vector being shorter = just re cycle
x <- c(1,2,3)
y <- c(3,4)
x+y = (4,6,6)
```

NumPy does this using broadcasting - find more on help page

Problem Set

- Counting number of species in the genus **(SELECT num_sp)** vs. counting number of species Greg has seen in that genus **(COUNT(species.species_ioc)**

genus_ioc	family_ioc	num_sp
Aulacorhynchus	Ramphastidae	11
Pteroglossus	Ramphastidae	14
Andigena	Ramphastidae	4
Ramphastos	Ramphastidae	8

genus	num_species
Andigena	1
Aulacorhynchus	2
Pteroglossus	4
Ramphastos	4


Happy
Thanksgiving!